AI DRIVEN MULTITRACK AUDIO MIXING USING FUZZY LOGIC TOOLS

by

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DEDICATION

This thesis is dedicated to my mother Alma Rosa, my father Luis Alfredo, my sister Marissa, my brother Wes, niece Lilly Rose, nephew Wesley Gunner, and to all of the people who have been in my life and shaped me into the person I have become.

Thank you for allowing me to dream.

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The art of audio mixing is an inherently subjective and nuanced process that requires years of experience to master. This thesis explores a novel use of AI to overcome some of the challenges of multitrack audio mixing for the non-expert. This thesis discusses data representation, common audio engineering practices, modeling of sounds into the stereo field, and proposes a solution for multitrack audio mixing using an artificial intelligent system incorporating fuzzy logic tools in Matlab. The artificial intelligent system is comprised of three modules that are each independently responsible for one of three dimensions in the sonic space. It is shown that by using fuzzy logic for the inherently fuzzy problem of audio mixing, a non-expert can achieve an improved mix of multiple audio tracks. For the purpose of testing, 8 multitrack songs consisting of varying instrumentation and styles were recorded and processed through the aforementioned AI system with improvements in the resultant mixes.

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ABBREVIATION

AI – Artificial Intelligence

ANN – Artificial Neural Network

DAW – Digital Audio Workstation

EQ – Equalization

FFT – Fast Fourier Transform

GUI – Graphical User Interface

HPF – High Pass Filter

LPF – Low Pass Filter

RBF – Radial Basis Function

CHAPTER ONE: INTRODUCTION

1.1 Background

The art of audio mixing in music is the process of manipulating recorded audio track parameters of multiple tracks to obtain a balanced, even distribution of sound. The goal is to blend multiple tracks into a single stereo mix that is appealing to the listener. A simple graphic demonstrating this is shown in Figure 1 below:

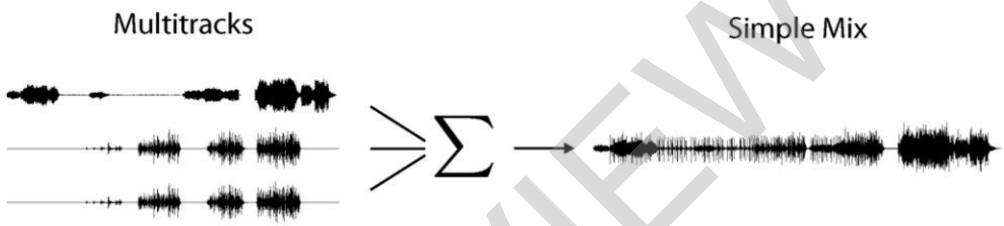

Figure 1: Diagram of multitrack audio being mixed into single mono track

Achieving an "appealing" mix is an inherently complex process that requires an expert-level understanding of the subtleties of audio mixing. As music and recording techniques have evolved over the last century, common practices in audio mixing have become apparent in the industry. For these reasons, there is a high learning-curve that non-experts must overcome to achieve a satisfactory mix. The challenge lies in knowing how to best adjust the parameters of tracks to optimize the interaction between the sonic energy content that, if left untouched, may result in a loss of clarity/definition in the final output mix yielding an unpleasant or unappealing listening experience. The use of an automatic mixing system addresses the needs of musicians who do not have the time, expertise, or inclination to perform the audio engineering required [15].

1.2 Visualizing Sound in 3D Space

In order to better understand some of the design choices that will be discussed in further chapters, it is useful to be able to think of these sounds in 3D space. There are essentially three main parameters that can be manipulated in a recorded audio track: equalization (EQ), volume, and stereo panning. With adjustments of these three parameters, audio engineers are able to blend multiple tracks into a more effective and/or aesthetically appealing mix.

We can model sounds into a visual framework by assigning the volume, panning, and equalization (EQ) parameters of a track as component values for coordinates in the x, y, and z space. We can imagine individual sounds in between a pair of stereo speakers as spheres in a 3D space where sounds are moved along the x-axis according to its panning (left speaker or right speaker) setting, moved along the y-axis according to its volume, and moved along the z-axis according to its frequency equalization. The size of the spheres are directly correlated to the intrinsic volumes of the tracks and their frequency content. Louder tracks will be perceived as larger objects in the 3D stereo space as they are "closer" to the listener, whereas quieter tracks are perceived as smaller objects and are "further away" from the listener. Tracks with lower frequency content are perceived to occupy more of the stereo space and are modeled with larger radii.

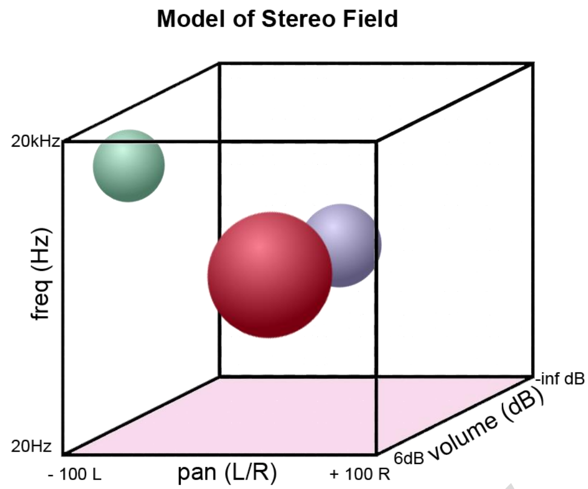

Figure 2: Diagram of stereo field with three tracks

We can see an example of this in figure 2. The various spheres represent different sounds (in this case tracks) that lie in at some base coordinate (x, y, z); in this scenario the red sphere (track 1) is louder than the lilac sphere (track 2), and is therefore modeled as being in front of the lilac sphere. The green sphere (track 3) is a track with high frequency information (perhaps cymbals, a glockenspiel, or wind chimes) and is also panned towards the left and is, therefore, modeled as being in the upper left corner of the stereo field. We bound the z -dimension of the stereo space by using the fact that the average human can only hear frequencies within a range from 20 Hz to 20 kHz. We bound the x - dimension by defining a pan all the way left as having a value of -100, a pan center as having a value of 0, and a pan all the way right as having a value of $+100$. Lastly, we can bound the y-dimension by defining no sound at all as $-inf\ dB$, and max sound output as $+6\ dB$. With these three parameters we can place any sound in our virtual stereo space. Now that we have a framework for visualizing our sounds, we can discuss some of the common practices followed in audio mixing which will be considered for the design of our AI mixer.

1.3 Audio Mixing Goals and Common Practices

1.3.1 EQ

The goal of EQ'ing is to shape our sounds to better fit into the sonic landscape of a mix of instruments and sounds. For example, if you have two similar sounds that occupy the same frequency range they will be competing with each other. In situations like this we can use EQ to allow both sounds to be heard in the mix. A general practice is using a LPF on sounds that were intended to serve compositionally as a bass instrument, and using a HPF on sounds that were not intended to serve as a bass instrument. When dealing with more complex EQ detailing, however, this is a particularly difficult parameter to objectify. EQing is fairly subjective and often varies based on musical genre and individual preference.

1.3.2 Volume

The goal of volume adjustment in mixing is to balance the sounds of multiple tracks. This is achieved by balancing the intensities of each track against each other. In order to create a balanced audio mixture, particular scaling of input gains and level faders must be achieved [10]. The general rule of thumb here is simply that high priority sounds should be at a louder volume, while lower priority sounds should be at a quieter volume. Compositionally speaking, it can be said that sounds intended to serve as the melody hold highest priority, followed by sounds intended to serve as the harmony, and lastly sounds intended to serve as the bass. Of course, the priority can vary depending on the genre of music and the preference of the audio engineer.